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# Research Statement

I am an applied macroeconomist, and my research interests are primarily in international economics and labor economics. My research is organized into two strands. The first leverages cutting-edge computational methods to analyze rich environments in international economics and finance with economically significant non-linearities and frictions. In parallel, I am interested in understanding how information shocks affect labor market outcomes through the combined use of granular data and modern advances in causal inference.

## 1 Sovereign Debt

### 1.1 How should governments borrow?

Most sovereign borrowing is denominated in a mix of the government's own currency and foreign currency, and a striking regularity in the data is that governments tend to issue domestic-currency (DC) debt at *shorter* maturities than foreign-currency (FC) debt. Standard quantitative models of sovereign debt, which hold one aspect constant, are silent on this fact. My job market paper, *"Everything in Moderation: The Currency-Maturity Composition of Sovereign Debt"* [1], builds the theory and computation needed to explain it.

My paper features an infinite-horizon quantitative sovereign default model in which the government chooses each period how much to borrow in short- and long-maturity bonds denominated in either domestic or foreign currency, whether to default, and how much to inflate. The resulting four-bond portfolio (short/long $\times$ DC/FC) lives in a six-dimensional state space that also tracks a persistent endowment process and the real exchange rate, which I estimate jointly for Türkiye. DC debt is distinctive in two respects: its real repayment value can be eroded by inflation, and since inflation is countercyclical, DC debt delivers state-contingent relief that FC debt cannot.

The model organizes the currency-maturity choice around a tension between *disciplining* and *hedging* benefits that holds separately along each margin, with the two margins compounding when combined. Along the currency margin, FC debt disciplines the government because lenders are never exposed to inflationary dilution, while DC debt hedges income risk because the government inflates when times are bad, so its real repayment value falls precisely when the marginal utility of consumption is high. The same trade-off recurs along the maturity margin, for a different reason: short-term debt disciplines because it carries no risk of being diluted by future re-borrowing, while long-term debt hedges because its market value—and so the real cost of repaying it—falls in exactly those bad states.

Critically, the two margins compound rather than offset. Issuing long-maturity DC debt raises lenders' expected future inflation by more than an equivalent amount of short-term DC debt, on top of the ordinary maturity-dilution risk already present in any long-term bond. Thus, long-maturity DC debt carries the worst incentive cost of all four instruments even as it carries the best hedge. The government avoids stacking both hedging channels onto a single instrument, which is why it borrows short in domestic currency and long in foreign currency rather than the reverse.

## 1.2 Computational methods for models of portfolio choice

Models of sovereign debt with multiple debt instruments often pose significant computational challenges, especially when it comes to accurately approximating the government's portfolio choice policy function. In *"Optimal Portfolio Uniqueness in Sovereign Debt Models"* [2] (joint with Zachary Stangebye), we explore various computational methods to robustly address these numerical difficulties. We document two primary obstacles. First, the domain over which the government's preferences over different borrowing strategies are well-defined is irregular. In states where the government defaults in equilibrium, the government is indifferent across its borrowing choices. Second, even in states where repayment is preferred, we find that standard parameterizations of the government's preferences yield near indifference across its portfolio choices.

We show that these numerical challenges can be overcome with the combined use of sophisticated function approximation techniques and Epstein–Zin preferences. Advanced interpolation strategies, such as Adaptive Sparse Grids and Gaussian Process Regression, can accommodate the irregularities in the government's portfolio choice domain since they can endogenously detect regions where the approximation is poorest and allocate more computational resources to those regions. Meanwhile, Epstein–Zin preferences mitigate the near-indifference problem by allowing for a separation between risk aversion and intertemporal substitution, which enhances the government's sensitivity to differences in portfolio choices.

# 2 Cross-Firm Pay Transparency and Firm Wage-Posting Behavior

In *"How Much Can I Make? Cross-Firm Pay Transparency's Effects on the US Labor Market"* [3], which I am writing with my colleague Suryansh Upmanyu, we estimate the causal effects of the recent wave of state-level cross-firm pay-transparency mandates (in Colorado, California, and Washington state) on local labor markets. We use difference-in-differences designs robust to staggered adoption to (i) quantify the extent to which the laws caused the amount of pay information to increase, (ii) estimate the effects of this increase on wages and employment, and (iii) explore how these effects vary across workers.

Using data on the near-universe of online vacancy postings across the U.S. from Lightcast, we estimate that these laws increased the fraction of postings with wage information by 24.7 percentage points, and that this increase was concentrated in higher-wage occupations. We

augment these first-stage results by using survey data on more than 3 million U.S. individuals from the American Community Survey to analyze how these laws affected real labor market outcomes. We find that the increase in pay transparency translated into average real wage increases of 3.2–4.4% for workers in Colorado and 0.6–1.3% in California and Washington state.

## 3 Future Research

In the coming years, I plan to further develop my research in both strands. For my research on pay transparency, my co-author and I have recently applied for administrative access to Census data from the Longitudinal Employer-Household Dynamics (LEHD) program, which will extend our analysis in two crucial ways. First, its size and granularity will allow us to obtain much more precise estimates of whether the laws achieved their policy objective of reducing the gender pay gap in treated states. Second, it will enable us to explore how these laws affected job flows, such as whether the new information caused workers to switch jobs more frequently, and whether firms responded by changing their hiring or firing behavior.

I also aim to extend my research on sovereign debt with many future projects. One project will leverage variation in institutional limits on state-level fiscal policy and taxes across U.S. states and quantitatively assess their costs and benefits. Additionally, I plan to quantify how declining population growth rates and aging populations affect future sovereign default risk. Lastly, I am interested in exploring how advanced iterative algorithms can be modified to solve sovereign debt models significantly more quickly and more accurately than standard value function iteration.

## References


[1] Raymond He. *"Everything in Moderation: The Currency-Maturity Composition of Sovereign Debt."* Job Market Paper. 2026. URL: https://raymondxuanhe.github.io/files/research/jmp/raymond_he_JMP_latest_version.pdf.

[2] Raymond He and Zachary Stangebye. *Optimal Portfolio Uniqueness in Sovereign Debt Models*. Working paper, 2026.

[3] Raymond He and Suryansh Upmanyu. *"How Much Can I Make? Cross-Firm Pay Transparency's Effects on the US Labor Market"*. SSRN Working Paper No. 6332818. 2026. DOI: 10.2139/ssrn.6332818. URL: https://dx.doi.org/10.2139/ssrn.6332818.